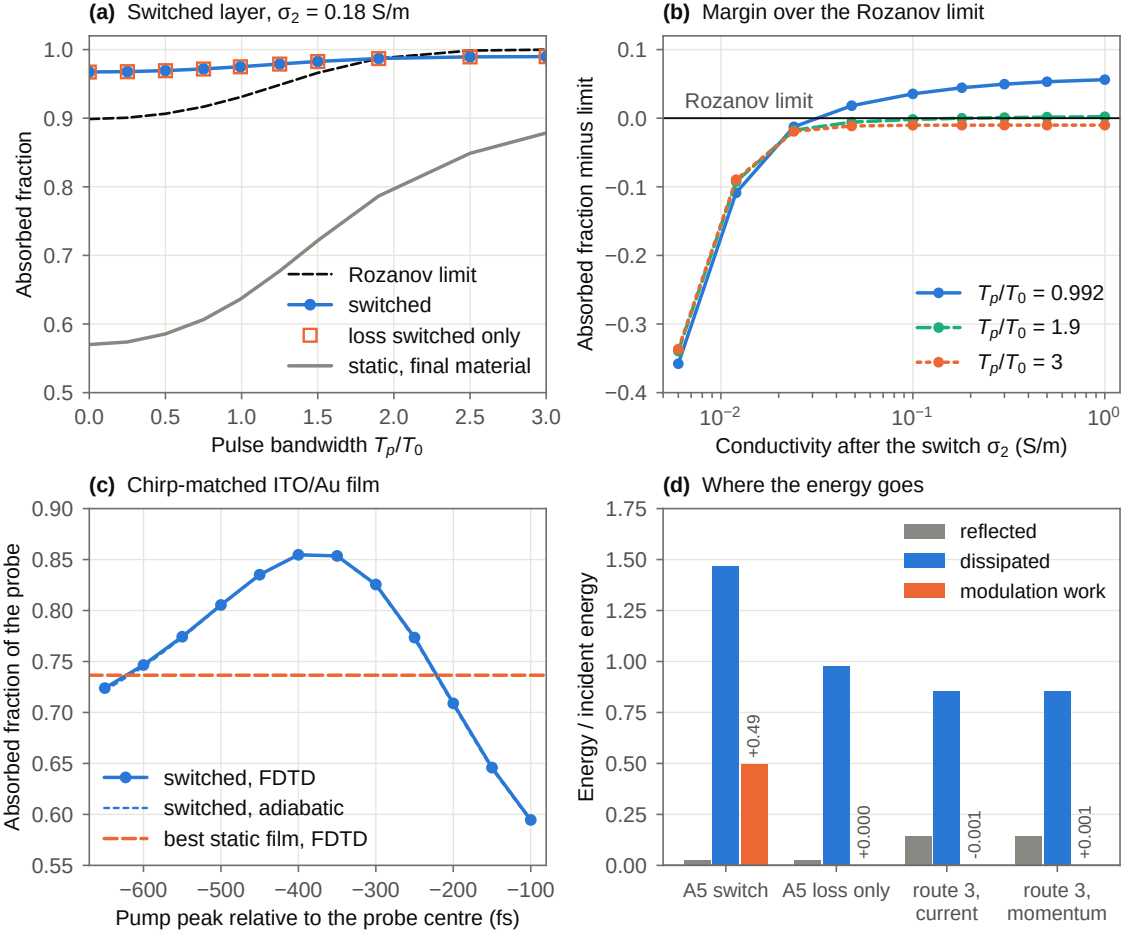
(a) Switched layer, σ2 = 0.18 S/m
Absorbed fraction
Pulse bandwidth Tp/T0
Rozanov limit
switched
loss switched only
static, final material
(b) Margin over the Rozanov limit
Absorbed fraction minus limit
Rozanov limit
Tp/T0 = 0.992
Tp/T0 = 1.9
Tp/T0 = 3
Conductivity after the switch σ2 (S/m)
(c) Chirp-matched ITO/Au film
Absorbed fraction of the probe
switched, FDTD
switched, adiabatic
best static film, FDTD
Pump peak relative to the probe centre (fs)
(d) Where the energy goes
Energy / incident energy
reflected
dissipated
modulation work
+0.49
+0.000
-0.001
+0.001
A5 switch
A5 loss only
route 3, current
route 3, momentum